ECO3121 Problem Set 3

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November 15, 2023

Question 1.

1. Here are some possible reasons why it could be biased:

   - **Omitted variables bias.** There are variables that influence both the household's decision to rent out land and its yield, but are not included in the regression model.

     The bias direction depends on the property of the omitted variable itself. For example, if richer families tend to rent out the field, and the yield is higher because of better resources, then it will cause the **upwards** of the causality. Conversely, if the field has a lower yield thus being rent out, it will cause a **downwards** of the causality.

   - **Simultaneity or reverse causality.** There could be a reverse causality where higher yields influence the decision to rent out land. Households experiencing high yields might be more confident in renting out their land, knowing that their remaining land is productive enough to meet their needs.

     This could lead to an **upwards** bias, as the model would capture the effect of high yields on renting behavior, not the other way around.

   - **Measurement Error.** If there is a measurement error in the independent variable, it could bias the results. This is particularly relevant in agricultural studies where precise measurement can be challenging.

     Typically, single measurement error in the independent variable leads to attenuation bias, meaning the estimated effect would lead to a **downwards** bias to zero.

2. This IV is **not true** after justification. Here are assumptions I check for:

   - **Check relevance:** The instrumental variable (total land area) must be correlated with the endogenous explanatory variable (rental out share).

     It is plausible as households with more land might be more inclined or able to rent out a portion of it.

   - **Check exogeneity:** The total land area must not be correlated with the error term in the regression model, and only influence the agricultural yield through its effect on the rental out share.

     This assumption fail to be reasonable. The total land area could have a direct effect on agricultural yield, independent of land rental activities. Larger areas might lead to economies of scale or more efficient use of resources influencing the yield.

3. First we use the Rainfall.dta, transform its *vl_id* to string to guarantee unity, and save it to the working directory.

```
use "/Users/kevinshuey/Github/coursework/eco3121/as3/Rainfall.dta", clear
tostring vl_id, replace
save Rainfall_temp.dta, replace
```

   After we prepare for the lyield:

```
use "/Users/kevinshuey/Github/coursework/eco3121/as3/aghousehold.dta", clear
gen yield=d32/d31
gen lyield = ln(yield)
```

we merge the datasets and drop the missing values.

```
merge m:1 vl_id using Rainfall_temp.dta
drop if missing(d31) | missing(av_rain) | missing(lyield)
```

Here we have the first stage regression model:

$$rental_out_share_i = \pi_1 av_rain + \pi_0$$

then we perform the regression

```
reg d31 av_rain
```

and the output

Source	SS	df	MS			
				Number of obs	=	13,862
				F(1, 13860)	=	5.70
Model	5932466.58	1	5932466.58	Prob > F	=	0.0170
Residual	1.4426e+10	13,860	1040804.56	R-squared	=	0.0004
				Adj R-squared	=	0.0003
Total	1.4431e+10	13,861	1041157.47	Root MSE	=	1020.2

d31	Coef.	Std. Err.	t	P>|t|	[95% Conf.	Interval]
av_rain	.0462179	.0193587	2.39	0.017	.0082721	.0841636
_cons	-23.16918	19.81693	-1.17	0.242	-62.01304	15.67468

The result indicates that one unit of increase in *av_rain* will lead to 0.04622 increase in the *rental_out_share*. We find the F-statistics to be 5.70 thus indicating a weak IV.

4. The 2nd stage IV point estimate is here as

$$lyield_i = \beta_0 + \beta_1 rental_out_share_hat_i + \mu_i$$

with the code as

```
predict rental_out_share_hat
reg lyield rental_out_share_hat
```

and the output

Source	SS	df	MS			
				Number of obs	=	13,862
				F(1, 13860)	=	4.84
Model	1.37754521	1	1.37754521	Prob > F	=	0.0279
Residual	3946.39339	13,860	.284732568	R-squared	=	0.0003
				Adj R-squared	=	0.0003
Total	3947.77094	13,861	.284811409	Root MSE	=	.5336

lyield	Coef.	Std. Err.	t	P>|t|	[95% Conf.	Interval]
rental_out_sha~t	.0004819	.0002191	2.20	0.028	.0000525	.0009113
_cons	5.93947	.0062102	956.40	0.000	5.927297	5.951643

The result indicates that one unit of increase in *rental_ out_ share_ hat* will lead to 0.0004819 increase in the *rental_ out_ share*.

We have the t-statistics to be

$$t = \frac{\hat{\beta}_1}{se(\hat{\beta}_1)} = \frac{0.0004819}{0.0002191} = 2.199 > 1.96$$

Thus we have 95% confidence to believe that there is causality between *rental_ out_ share_ hat* and *rental_ out_ share*.

5. With the direct command

```
ivreg lyield (d31 = av_rain)
```

we have

```
Instrumental variables (2SLS) regression

      Source |       SS           df       MS      Number of obs   =    13,862
-------------+----------------------------------   F(1, 13860)     =      2.20
       Model |  -4731.31401         1 -4731.31401   Prob > F        =    0.1380
    Residual |   8679.08494    13,860  .626196605   R-squared       =         .
-------------+----------------------------------   Adj R-squared   =         .
       Total |   3947.77094    13,861  .284811409   Root MSE        =    .79133

------------------------------------------------------------------------------
      lyield |      Coef.   Std. Err.      t    P>|t|     [95% Conf. Interval]
-------------+----------------------------------------------------------------
         d31 |   .0004819   .0003249     1.48   0.138     -.000155    .0011187
       _cons |    5.93947   .0092097   644.92   0.000     5.921418    5.957522
------------------------------------------------------------------------------
Instrumented:  d31
Instruments:   av_rain
------------------------------------------------------------------------------
```

We find β_{IV} to be the same, while the standard error seems larger in (5).

6. After executing the results:

```
use aghousehold_temp.dta, clear
drop _merge
gen vl_id2 = substr(vl_id, 1, 2)
merge m:1 vl_id2 using landlaw.dta
drop if missing(implemented) | missing(av_rain) | missing(lyield)
ivreg lyield (d31 = av_rain implemented)
```

we have the result

```
Instrumental variables (2SLS) regression

      Source |       SS           df       MS      Number of obs   =    13,862
-------------+----------------------------------   F(1, 13860)     =      0.58
       Model |  127.549923         1  127.549923   Prob > F        =    0.4462
    Residual |  3820.22102    13,860  .275629222   R-squared       =    0.0323
-------------+----------------------------------   Adj R-squared   =    0.0322
       Total |  3947.77094    13,861  .284811409   Root MSE        =      .525

------------------------------------------------------------------------------
      lyield |      Coef.   Std. Err.      t    P>|t|     [95% Conf. Interval]
-------------+----------------------------------------------------------------
         d31 |   -.000131    .000172    -0.76   0.446    -.0004681    .0002061
       _cons |   5.951348   .0055671  1069.01   0.000     5.940436    5.962261
------------------------------------------------------------------------------
Instrumented:  d31
Instruments:   av_rain implemented
------------------------------------------------------------------------------
```

Also, we could get comprehensive information about fisrt stage regresssion by executing

```
ivreg2 lyield (d31 = av_rain implemented), first
```

We begin to check for relevance and exogeneity:

- **Check relevance:** Here is part of the output mentioned above:

```
Sanderson-Windmeijer multivariate F test of excluded instruments:
  F(  2, 13859) =     4.48
  Prob > F      =   0.0114
```

As we have the F-statistics to be 4.48 which indicates a weak instrument, we decide to apply the Anderson-Rubin test. First we ensure the null hypothesis:

$$H_0 : \beta_1 = \beta_{1,IV} \qquad H_1 : \beta_1 \neq \beta_{1,IV}$$

then we compute

$$\epsilon_i = lyield_i - \beta_{1,IV}\, rental_out_share_i$$

Regress ϵ_i on $rental_out_share_i$, av_rain_i and $implemented_i$, and conduct F-test on the joint significance of the coefficients of av_rain and $implemented$. This process is shown as:

```
Weak-instrument-robust inference
Tests of joint significance of endogenous regressors B1 in main equation
Ho: B1=0 and orthogonality conditions are valid
Anderson-Rubin Wald test          F(2,13859)=    11.06     P-val=0.0000
Anderson-Rubin Wald test          Chi-sq(2)=     22.13     P-val=0.0000
Stock-Wright LM S statistic       Chi-sq(2)=     22.10     P-val=0.0000

Number of observations              N  =      13862
Number of regressors                K  =          2
Number of endogenous regressors     K1 =          1
Number of instruments               L  =          3
Number of excluded instruments      L1 =          2
```

We find the joint F-statistics to be 11.06, with p-value approaching 0. Hence we reject H_0, and there is significant relevance.

- **Check exogeneity:** As the single independent variable here is explained by 2 IVs, we apply Hansen-Sargan J test. First we obtain the 2SLS estimator with 2 IVs, but predict $lyield_estimate_i$ using only $rental_out_share_i$. Then we compute the residual

$$\hat{u}_i = lyield_i - lyield_estimate_i$$

Finally we regress $\hat{u}_i$ on $rental_out_share_i$, av_rain and $implemented_i$, and compute the F-statistic with the corresponding J-statistic:

$$J = 2F \sim \chi^2(1)$$

which tests the hypothesis that the coefficients on av_rain and $implemented_i$ are zero. Here are the details:

```
Sargan statistic (overidentification test of all instruments):          22.253
                                                   Chi-sq(1) P-val =    0.0000
```

We have the J-statistic to be 22.253, with p-value approaching 0. Thus there is significant exogeneity.

Appendix: Here is the .do File for Problem 1.

```
use "/Users/kevinshuey/Github/coursework/eco3121/as3/Rainfall.dta", clear
tostring vl_id, replace
save Rainfall_temp.dta, replace

use aghousehold.dta, clear
gen yield=d32/d31
gen lyield = ln(yield)
merge m:1 vl_id using Rainfall_temp.dta
drop if missing(d31) | missing(av_rain) | missing(lyield)
reg d31 av_rain
predict rental_out_share_hat
reg lyield rental_out_share_hat
ivreg lyield (d31 = av_rain)
save aghousehold_temp.dta

use aghousehold_temp.dta, clear
drop _merge
gen vl_id2 = substr(vl_id, 1, 2)
merge m:1 vl_id2 using landlaw.dta
drop if missing(implemented) | missing(av_rain) | missing(lyield)
ivreg lyield (d31 = av_rain implemented)

ivreg2 lyield (d31 = av_rain implemented), first
```

********** **This is the end of Problem Set 3.** **********